Highlights

A Multi-Layer Framework for Filtering, Merit Classification, Sentence Segmentation, and Named-Entity Extraction in Brazilian Criminal Labor-Law Decisions

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- Multi-layer filtering for noisy, heterogeneous legal corpora.
- Task-specific normalization improves classification and NER compatibility.
- Conviction-focused segmentation and legal-domain NER for structured extraction.
- Reproducible benchmark design for Portuguese criminal labor-law decisions.

A Multi-Layer Framework for Filtering, Merit Classification, Sentence Segmentation, and Named-Entity Extraction in Brazilian Criminal Labor-Law Decisions

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ABSTRACT

This paper presents a multi-layer legal NLP framework for extracting structured information from Brazilian criminal labor-law judicial decisions related to modern slavery. Starting from a heterogeneous corpus (2,581 documents), the framework applies: (i) document-type filtering to retain only final decisions, (ii) legal-text normalization with task-specific outputs for classification and NER, (iii) penal-merit classification using Legal-BERT embeddings with classical classifiers, (iv) LLM-assisted sentence segmentation for conviction-focused documents, and (v) legal-domain NER for entity extraction. We describe the full pipeline, evaluation protocol, and error analysis, and discuss practical deployment constraints, reproducibility, and ethical/legal considerations. Results indicate that layered filtering and task-oriented preprocessing substantially improve downstream extraction quality in low-resource legal settings.

1. Introduction

1.1. Problem context and motivation

- Explain the supply-chain transparency challenge in modern slavery investigations.
- Justify why judicial decisions are high-value evidence sources.

1.2. Research gap

- Existing legal NLP pipelines are often single-stage and not robust to document heterogeneity.
- Limited resources for Portuguese criminal labor-law datasets and benchmarks.

1.3. Research questions

- RQ1: How much does multi-layer filtering improve downstream extraction reliability?
- RQ2: Which merit-classification model is most robust under class imbalance?
- RQ3: How do segmentation quality and NER quality interact in the final extraction output?

1.4. Main contributions

1. End-to-end framework from raw legal documents to structured entities.
2. Task-adaptive preprocessing design with separate outputs for classification and NER.
3. Comparative evaluation of embedding-based classifiers for penal-merit prediction.
4. Practical protocol for LLM-assisted segmentation plus legal-domain NER.

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2. Related Work

2.1. Legal document filtering and classification

2.2. Legal sentence segmentation and structure mining

2.3. Legal NER in low-resource languages

2.4. Comparative synthesis with at least three high-impact studies

Required output in this section: a compact comparison table with dataset size, language, task formulation, model family, metrics, and limitations.

3. Dataset and Problem Definition

3.1. Corpus description

- Raw corpus size: 2,581 documents.
- Document heterogeneity (sentença, acórdão, despacho, ata, etc.).

3.2. Task definitions

- T1: Document-type filtering.
- T2: Penal-merit classification (classes: condenação, absolvição, extinto).
- T3: Decision segmentation.
- T4: Named-entity extraction.

3.3. Annotation and label governance

- Define legal criteria for label assignment (Art. 149 framing).
- Report annotation process and expert validation protocol.

3.4. Data governance and ethics

- LGPD/privacy handling, anonymization strategy, and legal-compliance statement.

4. Methodology

4.1. Overview of the multi-layer pipeline

Figure 1 presents the end-to-end workflow adopted in this study. The process starts with document-type normalization and filtering (Stage1_1), keeping only judicial sentences that are relevant to criminal-merit analysis. Next, Stage1_2 applies task-adaptive preprocessing and generates two normalized outputs: one optimized for merit classification and one that preserves legal-text patterns for segmentation and NER. Stage2 performs penal-merit classification, Stage3 applies LLM-assisted segmentation to the conviction subset, and Stage4 performs legal-domain named-entity extraction over segmented content.

4.2. Stage1_1: Document-type normalization and filtering

- Pattern-based extraction of document type from metadata-like spans.
- Final retention of *sentença* documents.

4.3. Stage1_2: Task-adaptive legal text preprocessing

- PJe duplicate-footer removal.
- Aggressive normalization for classification output.
- Conservative normalization for NER/segmentation output.

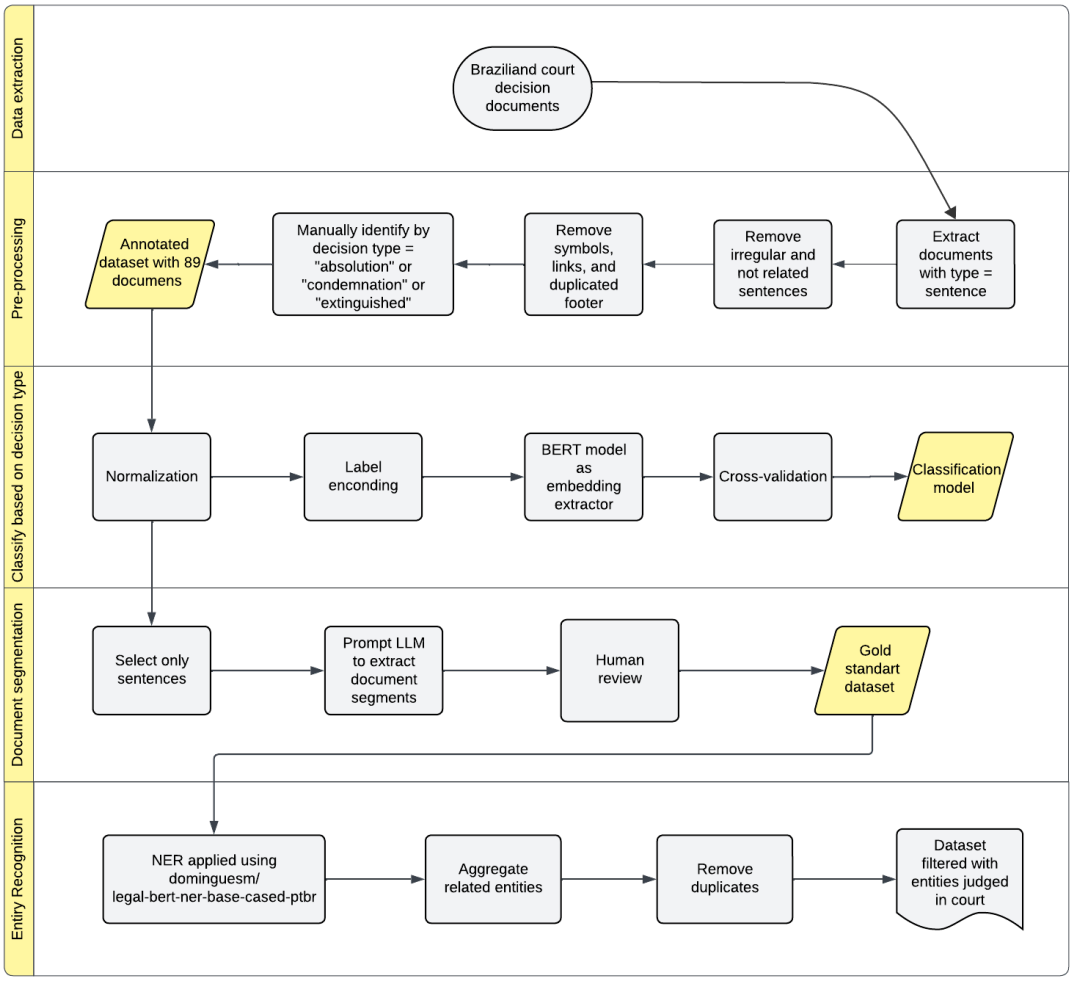

Figure 1: Activity diagram of the proposed multi-layer pipeline for legal document filtering, penal-merit classification, segmentation, and NER extraction.

4.4. Stage2: Penal-merit classification with Legal-BERT embeddings

- Encoders avaliados: `dominguesm/legal-bert-base-cased-ptbr` e `neuralmind/bert-base-portuguese-cased`.
- Classifiers: Logistic Regression, linear SVM, Random Forest, XGBoost.
- Stratified k-fold protocol and weighted metrics.

4.5. Stage3: LLM-assisted segmentation of conviction decisions

- Input restricted to conviction subset.
- Prompting protocol and JSON schema validation.
- Retry/failure handling and quality-control checks.

4.6. Stage4: Legal-domain NER extraction

- Model: `dominguesm/legal-bert-ner-base-cased-ptbr`.
- Entity aggregation strategy and output schema.

Table 1
Stage2 encoder comparison across downstream classifiers (weighted F1, mean ± std across folds).

Classifier	Legal-BERT (dominguesm)	BERTimbau (neuralmind)
XGBoost	$\mathbf{0.6114 \pm 0.0871}$	0.5261 ± 0.0590
SVM (Linear)	0.5415 ± 0.0484	0.5350 ± 0.0664
Random Forest	0.5406 ± 0.1249	0.5611 ± 0.0622
Logistic Regression	0.5399 ± 0.0522	0.5627 ± 0.0613

5. Experimental Setup

5.1. Evaluation protocol

- Data splits, fold strategy, seeds, and confidence intervals.
- Primary metrics: macro-F1, weighted-F1, precision, recall, per-class scores.

5.2. Baselines and ablations

- Baseline without multi-layer filtering.
- Baseline with single normalization stream.
- Ablation removing each stage (one-at-a-time).

5.3. Implementation details

- Software stack, hardware profile, runtime, and cost profile (LLM calls).

6. Results

6.1. Merit classification performance

Table 1 compares the two encoder backbones under the same downstream classifiers, using weighted F1 as the primary criterion. The best overall result is highlighted in bold and corresponds to Legal-BERT with XGBoost ($F1 = 0.6114 \pm 0.0871$). BERTimbau shows small gains for Logistic Regression and Random Forest, but does not surpass the best Legal-BERT configuration.

In practical terms, this indicates that domain adaptation in the encoder is beneficial when paired with a stronger non-linear classifier (XGBoost), whereas gains are less consistent for simpler classifier families. Given this trade-off, we keep the legal encoder as the default configuration in Stage2 because it provides the highest best-case performance for the penal-merit task.

6.2. Segmentation quality

Report schema-valid rate, section-level agreement, and representative errors.

6.3. NER extraction quality

Report entity-level precision/recall/F1 (strict and relaxed matching if possible).

6.4. End-to-end pipeline gains

Compare extraction quality before vs. after multi-layer filtering.

7. Discussion

7.1. Why the pipeline works

7.2. Failure modes and error taxonomy

7.3. Generalization to other legal domains/languages

7.4. Operational and policy implications

8. Threats to Validity

- Internal validity: label noise, annotation drift, prompt sensitivity.

- External validity: transfer to other courts, years, and jurisdictions.
- Construct validity: legal meaning vs. NLP proxy metrics.

9. Reproducibility Statement

- Public code structure, dependency pinning, random seeds, and artifact inventory.
- Which resources can/cannot be publicly shared due to legal constraints.

10. Conclusion and Future Work

- Summarize empirical and practical contributions.
- Future work: active learning, expert-in-the-loop verification, and cross-court transfer evaluation.

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CRediT authorship contribution statement

Carlos A. V. Tinin: Conceptualization, Methodology, Software, Writing – original draft. **Edison P. de Freitas:** Validation, Formal analysis, Writing – review & editing. **Kevin H. M. Gularte:** Validation, Formal analysis, Writing – review & editing. **Elena J. da Costa:** Validation, Formal analysis, Writing – review & editing. **João Paulo J. da Costa:** Validation, Formal analysis, Writing – review & editing.